

UNITED REPUBLIC OF TANZANIA

PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM SIX - PRE MOCK EXAMINATION

143/2

MATHEMATICS 2

PROPOSED MARKING SCHEME

wazaelimu.com

August 2026

Page 1

1 (a)

Soln.

Page 02

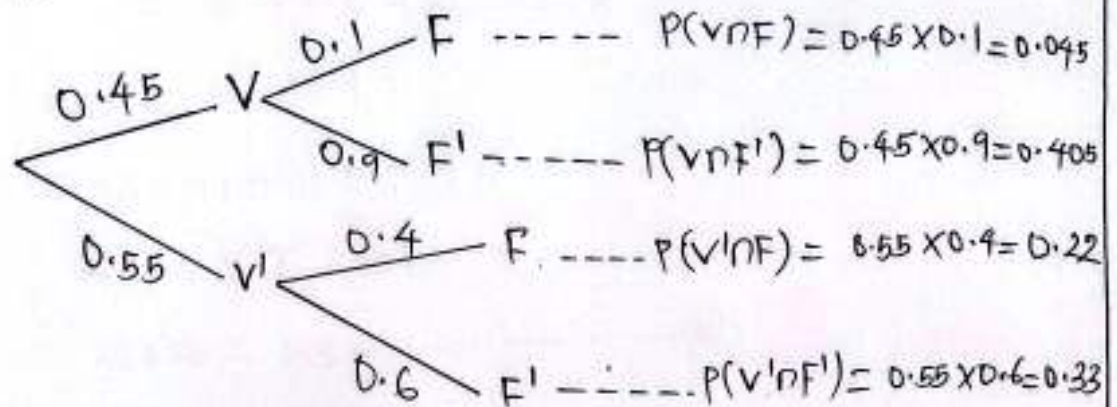
let V be elderly people who have vaccination
 V' be elderly people who have no vaccination
 F elderly people with flu
 F' elderly people with no flu.

Given that

$$P(V) = 45\% = 0.45$$

$$P(V') = 1 - 45\% = 0.55$$

Consider the tree diagram below



(a)(i) Required to find the probability that an elderly person chosen at random gets flu is

$$P(F) = P(V \cap F) + P(V' \cap F)$$

$$P(F) = 0.045 + 0.22 = 0.265$$

0.265

$\therefore$ Probability that an elderly person chosen at random gets flu is 0.265

$$(a)(ii) P(V/F) = \frac{P(V \cap F)}{P(F)} = \frac{0.045}{0.265} = 0.17578125$$

0.176

Probability that an elderly person chosen at random had the vaccinated on, given that they get flu is 0.176.

1 (b)

Soln.

Page 03

Given

$$E(x) = 2.7$$

Required the value of 'a' and 'b'

$$a + 0.3 + b + 0.2 = 1$$

$$a + b + 0.5 = 1$$

$$a + b = 1 - 0.5$$

$$a + b = 0.5 \text{ ————— (i) } \textcircled{01}$$

$$E(x) = 1(a) + 2(0.3) + 3b + 4(0.2)$$

$$= a + 0.6 + 3b + 0.8$$

$$E(x) = a + 3b + 1.4$$

$$a + 3b + 1.4 = 2.7$$

$$a + 3b = 2.7 - 1.4$$

$$a + 3b = 1.3 \text{ ————— (ii) } \textcircled{01}$$

$$a = 0.1, b = 0.4 \text{ } \textcircled{01}$$

(c)(i)

Soln.

Given

$$f(x) = \begin{cases} \frac{x}{6}, & \text{for } 0 \leq x < 3 \\ \frac{4-x}{2}, & \text{for } 3 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

for $0 \leq x < 3 = \frac{x}{6}$ This is a linear function

$$\text{At } x = 0$$

$$\Rightarrow f(0) = 0 \text{ } \textcircled{01} \text{ } \textcircled{01} \frac{1}{2}$$

As x approaches 3 from left $f(x)$ approaches $\frac{3}{6} = \frac{1}{2}$

1c (i) for $3 \leq x \leq 4$

$$\varphi(x) = \frac{4-x}{2}$$

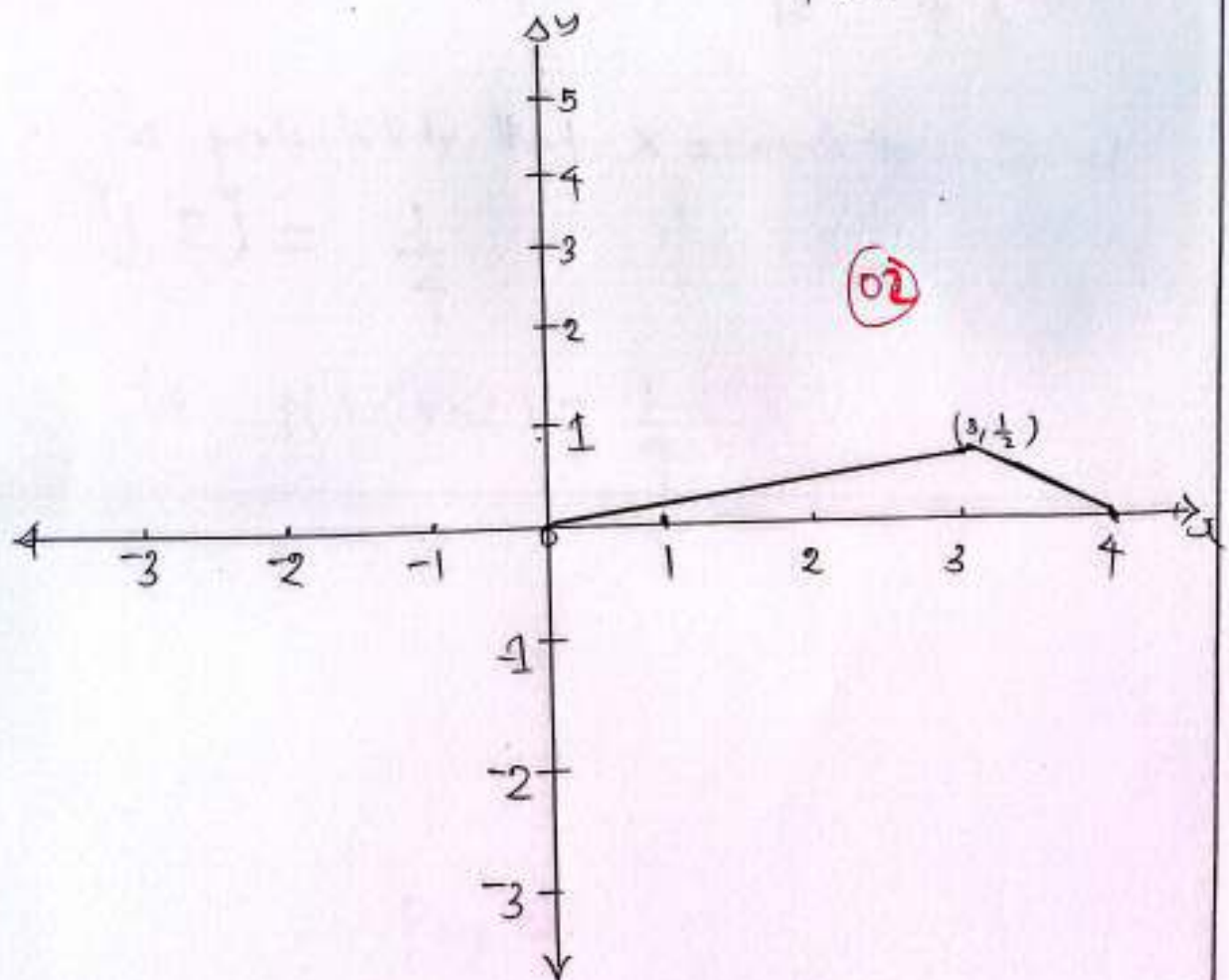
$$\text{At } x=3 \Rightarrow \varphi(3) = \frac{1}{2} \text{ (01)}$$

$$\text{At } x=4 \Rightarrow \frac{4-4}{2} = 0$$

$$\Rightarrow \varphi(4) = 0$$

So we have points $(3, \frac{1}{2})$ and $(4, 0)$ (01)

$\varphi(x) = 0$ otherwise implies that graph is on the X-axis for all values of x .



1. Q(ii) Required to find the probability that x occurs in the interval $[1, 2]$.

ie

$$P(a \leq x \leq b) = \int_a^b f(x) dx$$

where $[a, b] = [1, 2]$

$$P(1 \leq x \leq 2) = \int_1^2 \frac{x}{6} dx = \left[\frac{x^2}{12} \right]_1^2 \quad (01)$$

$$P(1 \leq x \leq 2) = \int_1^2 \frac{x}{6} dx = \frac{3}{12} = \frac{1}{4} \quad (01)$$

∴ The probability that x occurs in the interval $[1, 2] = \frac{1}{4}$

ie $P(1 \leq x \leq 2) = \frac{1}{4} \quad (01)$

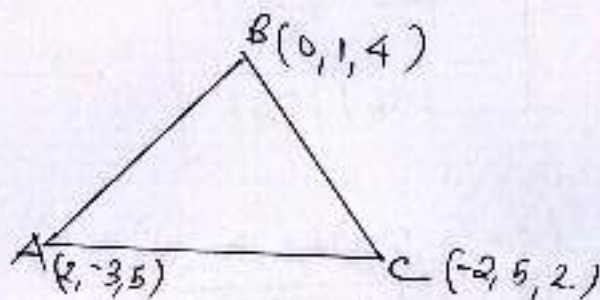
2 (a)

Soln.

Page 06

Given the vertices of ΔABC as

$A(2, -3, 5)$, $B(0, 1, 4)$ and $C(-2, 5, 2)$
 required to find the angles of a triangle



Angle $\hat{A}$ is between $\vec{AB}$ and $\vec{AC}$
 but

$$\vec{AB} = (0, 1, 4) - (2, -3, 5) = (-2, 4, 1) \quad \text{OOI} \frac{1}{2}$$

$$\vec{AC} = (-2, 5, 2) - (2, -3, 5) = (-4, 8, -3) \quad \text{OOI} \frac{1}{2}$$

$$\hat{A} = \cos^{-1} \left[\frac{\vec{AB} \cdot \vec{AC}}{|\vec{AB}| |\vec{AC}|} \right] \quad \text{OOI} \frac{1}{2}$$

$$\hat{A} = \cos^{-1} \left[\frac{(-2, 4, 1) \cdot (-4, 8, -3)}{(\sqrt{(-2)^2 + 4^2 + 1^2})(\sqrt{(-4)^2 + 8^2 + (-3)^2})} \right] \quad \text{OOI} \frac{1}{2}$$

$$\hat{A} = \cos^{-1} \left[\frac{8 + 32 + 3}{\sqrt{21} \times \sqrt{89}} \right]$$

$$\hat{A} = \cos^{-1} \left(\frac{43}{\sqrt{1869}} \right)$$

$$\therefore \hat{A} = 5.937595316 \approx 5.94^\circ = 5.56' \quad \text{OOI} \frac{1}{2}$$

2 (a) Angle B is between $\vec{BA}$ and $\vec{BC}$

$$\vec{BA} = (2, -3, 5) - (0, 1, 4) = (2, -4, 1)$$

$$\vec{BC} = (2, 5, 2) - (0, 1, 4) = (-2, 4, -2)$$

from

$$\hat{B} = \cos^{-1} \left[\frac{\vec{BA} \cdot \vec{BC}}{|\vec{BA}| |\vec{BC}|} \right] \quad \text{001 } \frac{1}{2}$$

$$\hat{B} = \cos^{-1} \left[\frac{(2, -4, 1) \cdot (-2, 4, -2)}{(\sqrt{(2)^2 + (-4)^2 + (1)^2})(\sqrt{(-2)^2 + (4)^2 + (-2)^2})} \right]$$

$$\hat{B} = \cos^{-1} \left(\frac{-22}{\sqrt{504}} \right)$$

$$\therefore \hat{B} = 168.51^\circ \approx 168^\circ 30' \quad \text{001 } \frac{1}{2}$$

Angle $\hat{C}$ is between $\vec{CA}$ and $\vec{CB}$

$$\vec{CA} = (2, -3, 5) - (-2, 5, 2) = (4, -8, 3)$$

$$\vec{CB} = (0, 1, 4) - (-2, 5, 2) = (2, -4, 2)$$

Then

$$\hat{C} = \cos^{-1} \left[\frac{\vec{CA} \cdot \vec{CB}}{|\vec{CA}| |\vec{CB}|} \right] = \cos^{-1} \left(\frac{46}{\sqrt{2136}} \right)$$

$$\hat{C} = \underline{5.55^\circ \approx 5^\circ 33'} \quad \text{001 } \frac{1}{2}$$

$\therefore$ The angles of a triangle ABC are 5.94° , 168.51° and 5.55° respectively (Q1)

2 (b)

Soln

Page 08

Given that

$$|a| = 2, |b| = 3, \theta = 60^\circ$$

Required to find $|a + 2b|$

from

$$|a + 2b|^2 = (a + 2b)^2 \quad (01)$$

$$a^2 + 4a \cdot b + 4b^2 = |a + 2b|^2$$

$$|a + 2b|^2 = a^2 + 4a \cdot b + 4b^2 \quad (07)$$

but

$$|a|^2 = a^2 = a \cdot a, \quad b^2 = b \cdot b = |b|^2, \quad a \cdot b = |a||b|\cos\theta$$

$$|a + 2b|^2 = |a|^2 + 4|a||b|\cos\theta + 4|b|^2 \quad (07)$$

$$= (2)^2 + 4(2)(3)\cos 60^\circ + 4(3)^2$$

$$= 4 + 24\cos 60^\circ + 36$$

$$= 40 + 24\cos 60^\circ \quad (07)$$

$$= 40 + 24\left(\frac{1}{2}\right)$$

$$= 40 + 12$$

$$|a + 2b| = 52$$

$$\therefore |a + 2b| = 52 \quad (01)$$

2 (c)

Soln

Page 09

The force of magnitude 5 units, acting in the direction of vector $6\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ is given by

$$\underline{F}_1 = 5 \left(\frac{6\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}}{\sqrt{6^2 + 2^2 + 3^2}} \right) = \frac{5}{7} (6\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}) \quad (61)$$

Also

$$\underline{F}_2 = 3 \left(\frac{3\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}}{\sqrt{3^2 + (-2)^2 + 6^2}} \right) = \frac{3}{7} (3\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}) \quad (62)$$

The resultant force is given by

$$\begin{aligned} \underline{F} &= \underline{F}_1 + \underline{F}_2 = \frac{5}{7} (6\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}) + \frac{3}{7} (3\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}) \\ &= \frac{1}{7} (30\mathbf{i} + 10\mathbf{j} + 15\mathbf{k} + 9\mathbf{i} - 6\mathbf{j} + 18\mathbf{k}) \end{aligned}$$

$$\Rightarrow \underline{F} = \frac{1}{7} (39\mathbf{i} + 4\mathbf{j} + 33\mathbf{k}) \quad (63)$$

displacement from $(2, 1, -3)$ to $(4, -3, 1)$ is given by

$$\underline{d} = (4\mathbf{i} - 3\mathbf{j} + \mathbf{k}) - (2\mathbf{i} + \mathbf{j} - 3\mathbf{k})$$

$$\underline{d} = 2\mathbf{i} - 4\mathbf{j} + 4\mathbf{k} \quad (64)$$

Required work done

$$W = \underline{F} \cdot \underline{d} = \left(\frac{1}{7} (39\mathbf{i} + 4\mathbf{j} + 33\mathbf{k}) \right) \cdot (2\mathbf{i} - 4\mathbf{j} + 4\mathbf{k})$$

$$\text{Work done} = \frac{194}{7} \text{ units} \approx 27.7 \text{ units} \quad (65)$$

(a) Given the equation

$$x^3 + ax^2 + bx + 30 = 0$$

and the roots of the equation are in the ratio 2:3:5

let k be the factor of the roots then

α , β and γ be roots of the equation

$$\Rightarrow \alpha = 2k, \beta = 3k \text{ and } \gamma = 5k, \text{ but (01)}$$

$$\text{From } x^3 + ax^2 + bx + 30 = 0$$

$$\alpha + \beta + \gamma = -a$$

$$\Rightarrow 2k + 3k + 5k = -a$$

$$10k = -a \text{ ————— (1) (01)}$$

Also

$$\alpha\beta + \alpha\gamma + \beta\gamma = b$$

$$\Rightarrow 6k^2 + 10k^2 + 15k^2 = b$$

$$31k^2 = b \text{ ————— (11) (01)}$$

$$\alpha\beta\gamma = -30$$

$$30k^3 = -30$$

$$\Rightarrow k = -1 \text{ (01)}$$

substituting $k = -1$ into eqn (1) and (11)

$$\Rightarrow a = 10 \text{ and}$$

$$\Rightarrow b = 31 \text{ (01)}$$

3 (b)

soln

Page 11

let / Given $f(x) = \frac{10 - 17x + 14x^2}{(2+x)(1-2x)^2}$

Required to partialize it.

$$\frac{10 - 17x + 14x^2}{(2-x)(1-2x)^2} = \frac{A}{2-x} + \frac{B}{1-2x} + \frac{C}{(1-2x)^2} \quad \text{⑥} \frac{1}{2}$$

$$\frac{10 - 17x + 14x^2}{(2-x)(1-2x)^2} = \frac{A(1-2x)^2 + B(2+x)(1-2x) + C(2+x)}{(2-x)(1-2x)^2} \quad \text{⑦} \frac{1}{2}$$

$$10 - 17x + 14x^2 = A(1-2x)^2 + B(2+x)(1-2x) + C(2+x)$$

let $x = -2$

$$100 = 25A \Rightarrow A = 4 \quad \text{⑧} \frac{1}{2}$$

let $x = \frac{1}{2}$

$$5 = \frac{5}{2}C \Rightarrow C = 2$$

let $x = 0$

$$10 = A + 2B + 2C$$

$$\Rightarrow 4 + 4 + 2B = 10$$

$$2B = 2$$

$$\Rightarrow B = 1$$

$$\therefore A = 4, B = 1 \text{ and } C = 2 \quad \text{⑨} \frac{1}{2}$$

$$\Rightarrow \frac{10 - 17x + 14x^2}{(2-x)(1-2x)^2} = \frac{4}{2-x} + \frac{1}{1-2x} + \frac{2}{(1-2x)^2} \quad \text{⑩} \frac{1}{2}$$

3(b) Now required to find the expansion up to and including the term in x^3 .

Consider

$$\frac{4}{2+x} + \frac{1}{1-2x} + \frac{2}{(1-2x)^2} = 4(2+x)^{-1} + 1(1-2x)^{-1} + 2(1-2x)^{-2} \quad \text{--- (i) } \frac{001}{2}$$

from

$$(1+x)^n = 1 + \frac{nx}{1!} + \frac{n(n-1)x^2}{2!} + \dots \quad \frac{001}{2}$$

$$(2+x)^{-1} = 2^{-1} \left(1 + \frac{x}{2}\right)^{-1} = \frac{1}{2} \left(1 + \frac{x}{2}\right)^{-1}$$

$$4(2+x)^{-1} = 4 \left(\frac{1}{2} - \frac{x}{4} + \frac{1}{8}x^2 - \frac{1}{16}x^3 + \dots \right) \quad \text{--- (ii) } \frac{001}{2}$$

$$(1-2x)^{-1} \stackrel{\text{A1n}}{=} 1 + 2x + 4x^2 + 8x^3 + \dots \quad \text{--- (iii) } \frac{001}{2}$$

Again

$$2(1-2x)^{-2} = 2(1 + 4x + 12x^2 + 32x^3 + \dots) \quad \text{--- (iv) } \frac{001}{2}$$

Adding eqn (i), (ii) and (iii)

$$\frac{4}{2+x} + \frac{1}{1-2x} + \frac{2}{(1-2x)^2} = 5 + 9x + 30x^2 + 68x^3 + \dots \quad \frac{001}{2}$$

$\therefore$ The expansion of $\frac{4}{2+x} + \frac{1}{1-2x} + \frac{2}{(1-2x)^2}$ is
 $5 + 9x + 30x^2 + 68x^3 + \dots$ $\frac{001}{2}$

3 (ci) Given

$$A = \begin{pmatrix} 2 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 1 & 3 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 & 1 & 2 \\ 3 & 2 & 2 \\ 1 & 0 & 1 \end{pmatrix}$$

required $|AB|$. but

$$AB = \begin{pmatrix} 2 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 3 & 1 & 2 \\ 3 & 2 & 2 \\ 1 & 0 & 1 \end{pmatrix}$$

$$AB = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{001 } \frac{1}{2}$$

$$|AB| = 1 \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} - 0 \begin{vmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} + 0 \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{vmatrix}$$

$$|AB| = 1. \quad \text{001 } \frac{1}{2}$$

(ii) Now Required to use determinant $|AB|$ to solve

$$2x - y - 2z = 6$$

$$-x + y + 0z = 1$$

$$-2x + y + 3z = -7$$

$$\begin{pmatrix} 2 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 6 \\ 1 \\ -7 \end{pmatrix} \quad \text{001 } \frac{1}{2}$$

$$X \text{ value} = \frac{\begin{vmatrix} 6 & -1 & -2 \\ 1 & 1 & 0 \\ -7 & 1 & 3 \end{vmatrix}}{|AB|} = \frac{5}{1} = 5 \quad \text{001 } \frac{1}{2}$$

3 (ii) for y value

$$y = \frac{\begin{vmatrix} 2 & 6 & -2 \\ -1 & 1 & 0 \\ -2 & -7 & 3 \end{vmatrix}}{|AB|} = \frac{6}{1} = 6 \quad \text{or } \frac{6}{2}$$

for z value

$$Z = \frac{\begin{vmatrix} 2 & -1 & 6 \\ -1 & 1 & 1 \\ -2 & 1 & -7 \end{vmatrix}}{|AB|} = \frac{-1}{1} = -1 \quad \text{or } \frac{6}{2}$$

∴ The solution of the simultaneous equations

$$\begin{cases} 2x - y - 2z = 6 \\ -x + y = 1 \\ -2x + y + 3 = -7 \end{cases}$$

are

$$x = 5, y = 6 \text{ and } z = -1 \quad \text{or}$$

Soln.

Page 15

4 (a) (i) $Z = x + iy$

Required to find the cartesian if $\arg(Z+i) = \frac{\pi}{4}$

$$Z = x + iy$$

$$\arg(Z+i) = \frac{\pi}{4}$$

$$\arg(x + iy + i) = \frac{\pi}{4}$$

$$\arg(x + i(1+y)) = \frac{\pi}{4}$$

$$\arg(Z) = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\tan^{-1}\left(\frac{1+y}{x}\right) = \frac{\pi}{4}$$

Apply tan both sides

$$\tan \tan^{-1}\left(\frac{1+y}{x}\right) = \tan \frac{\pi}{4}$$

$$\frac{1+y}{x} = 1$$

$$x = 1+y$$

$$x - y - 1 = 0$$

02 $\frac{1}{2}$

(a) (ii) Given Soln.

$$Z = a + ib$$

$$Z^5 = 1$$

Required to show that $4a(b^4 - a^4) = 1$

$$(a+ib)^5 = 1$$

By using Pascal triangle for expansion of $(a+b)^5$

$$\begin{array}{cccccc} & & 1 & & 2 & & 1 \\ & 1 & & 3 & & 3 & & 1 \\ 1 & 4 & & 6 & & 4 & & 1 \\ 1 & 5 & & 10 & & 10 & & 5 & & 1 \end{array}$$

01 $\frac{1}{2}$

$$4 \text{ aii)} \Rightarrow (a+ib)^5 = a^5 + 5a^4(ib) + 10a^3(ib)^2 + 10a^2(ib)^3 + 5a(ib)^4 + (ib)^5$$

$$\Rightarrow (a+ib)^5 = a^5 + 5a^4bi + 10a^3b^2i^2 + 10a^2b^3i^3 + 5ab^4i^4 + i^5b^5$$

$$\text{since } i^2 = -1, \quad i^3 = -i, \quad i^4 = 1 \text{ and } i^5 = i$$

$$\Rightarrow (a+ib)^5 = a^5 + 5a^4bi - 10a^3b^2 - 10ia^2b^3 + 5ab^4 + ib^5$$

grouping real part and imaginary part

$$\Rightarrow (a+ib)^5 = (a^5 - 10a^3b^2 + 5ab^4) + i(5a^4b - 10a^2b^3 + b^5)$$

since we have been given that $\frac{\theta}{2}$

$$Z^5 = 1 + 0i$$

$$\Rightarrow a^5 - 10a^3b^2 + 5ab^4 = 1 \quad \text{--- (i)} \quad \frac{\theta}{2}$$

$$5a^4b - 10a^2b^3 + b^5 = 0 \quad \text{--- (ii)}$$

from eqn (ii)

$$5a^4b - 10a^2b^3 + b^5 = 0$$

$$b(5a^4 - 10a^2b^2 + b^4) = 0$$

divide by b both sides

$$5a^4 - 10a^2b^2 + b^4 = 0$$

$$10a^2b^2 = 5a^4 + b^4 \quad \text{--- (iii)} \quad \frac{\theta}{2}$$

substitute eqn (iii) into (i)

$$a^5 - a(10a^2b^2) + 5ab^4 = 1$$

$$a^5 - a(5a^4 + b^4) + 5ab^4 = 1$$

$$a^5 - 5a^5 + 5ab^4 + 5ab^4 = 1$$

$$-4a^5 + 10ab^4 = 1$$

$$4a(b^4 - a^4) = 1 \quad \frac{\theta}{2}$$

Hence shown as required.

(2 1/2)

4 (b)

Soln

Page 17

Required to use de-Moivre's theorem to prove that

$$\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$$

Consider LHS

$$\tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta}$$

001/2

from

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta) \quad 001/2$$

$$\cos(4\theta) + i \sin 4\theta = (\cos \theta + i \sin \theta)^4$$

$$= \cos^4 \theta + 4 \cos^3 \theta (i \sin \theta) + 6 \cos^2 \theta (i \sin \theta)^2 + 4 \cos \theta (i \sin \theta)^3 + (i \sin \theta)^4$$

$$= \cos^4 \theta + 4i \cos^3 \theta \sin \theta + 6i^2 \cos^2 \theta \sin^2 \theta + 4i^3 \cos \theta \sin^3 \theta + i^4 \sin^4 \theta$$

$$= (\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta) + i(4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta)$$

$$= \cos^4 \theta - 6 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2 + i(4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta)$$

$$\cos 4\theta + i \sin 4\theta = (\cos^4 \theta - 6 \cos^2 \theta + 6 \cos^4 \theta + 1 - 2 \cos^2 \theta + \cos^4 \theta) + i(4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta) \quad (i)$$

Compare Real part and Imaginary part,

$$\sin 4\theta = 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta \quad \text{--- (i)}$$

$$\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1 \quad \text{--- (ii)} \quad 001/2$$

Divide eqn (i) by (ii)

$$\frac{\sin 4\theta}{\cos 4\theta} = \frac{4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta}{8 \cos^4 \theta - 8 \cos^2 \theta + 1} \quad 001/2$$

Dividing RHS both numerator and denominator by $\cos^4 \theta$

4 (b)

$$\tan 4\theta = \frac{4 \cos^3 \theta \sin \theta}{\cos 4\theta} - \frac{4 \cos \theta \sin^3 \theta}{\cos 4\theta}$$

$$\frac{8 \cos^4 \theta}{\cos 4\theta} - \frac{8 \cos^2 \theta}{\cos 4\theta} + \frac{1}{\cos 4\theta} \quad \text{001} \frac{1}{2}$$

$$= \frac{4 \tan \theta - 4 \tan^3 \theta}{8 - 8 \sec^2 \theta + \sec 4\theta}$$

$$= \frac{4 \tan \theta - 4 \tan^3 \theta}{8 - 8 \sec^2 \theta + (\sec^2 \theta)^2}$$

$$= \frac{4 \tan \theta - 4 \tan^3 \theta}{8 - 8 \sec^2 \theta + (1 + \tan^2 \theta)^2} \quad \text{001} \frac{1}{2}$$

$$= \frac{4 \tan \theta - 4 \tan^3 \theta}{8 - 8(1 + \tan^2 \theta) + (1 + 2 \tan^2 \theta + \tan^4 \theta)}$$

$$= \frac{4 \tan \theta - 4 \tan^3 \theta}{8 - 8 - 8 \tan^2 \theta + 1 + 2 \tan^2 \theta + \tan^4 \theta} \quad \text{001} \frac{1}{2}$$

$$\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta} \quad \text{001} \frac{1}{2}$$

hence proved

page (18)

4 (c)

soln.

Page 19

Required to solve the equation $z^5 - 3i + 3 = 0$

$$z^5 = -3 + 3i$$

Consider the complex $z = -3 + 3i$

$$|z| = r = \sqrt{x^2 + y^2}$$

$$= r = \sqrt{(-3)^2 + 3^2}$$

$$= r = \sqrt{9 + 9}$$

$$= r = \sqrt{18}$$

$$|z| = r = 3\sqrt{2} \quad \text{sol } \frac{1}{2}$$

$$\arg(z) = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{3}{-3}\right) = \tan^{-1}(-1)$$

$$\arg(z) = \theta = -\frac{\pi}{4} \quad \text{sol } \frac{1}{2}$$

from

$$z_{k+1} = r^{\frac{1}{n}} \left[\cos\left(\frac{2\pi k + \theta}{n}\right) + i \sin\left(\frac{2\pi k + \theta}{n}\right) \right] \quad \text{sol } \frac{1}{2}$$

where $k = 0, 1, 2, 3, 4$.

$$n = 5$$

$$\text{for } k=0; \quad z_1 = (3\sqrt{2})^{\frac{1}{5}} \left[\cos\left(\frac{2\pi(0) - \pi/4}{5}\right) + i \sin\left(\frac{2\pi(0) - \pi/4}{5}\right) \right]$$

$$k=0 \Rightarrow z_1 = (\sqrt{18})^{\frac{1}{5}} \left[\cos \frac{\pi}{20} - i \sin \frac{\pi}{20} \right] \quad \text{sol } \frac{1}{2}$$

$$k=1 \Rightarrow z_2 = (\sqrt{18})^{\frac{1}{5}} \left[\cos \frac{3\pi}{20} + i \sin \frac{3\pi}{20} \right]$$

$$k=2 \Rightarrow z_3 = (\sqrt{18})^{\frac{1}{5}} \left[\cos \frac{5\pi}{20} + i \sin \frac{5\pi}{20} \right]$$

4 c) for $k=3$

$$Z_4 = (\sqrt{18})^{\frac{1}{5}} \left[\cos \frac{23\pi}{20} + i \sin \frac{23\pi}{20} \right]$$

for $k=4$

$$Z_5 = (\sqrt{18})^{\frac{1}{5}} \left[\cos \frac{31\pi}{20} + i \sin \frac{31\pi}{20} \right] \text{ ok } \frac{1}{2}$$

$\therefore$ The roots of the equation $z^5 - 3i + 3 = 0$ are

$$Z_1 = (\sqrt{18})^{\frac{1}{5}} \left[\cos \frac{\pi}{20} - i \sin \frac{\pi}{20} \right] \text{ ok } \frac{1}{2}$$

$$Z_2 = (\sqrt{18})^{\frac{1}{5}} \left[\cos \frac{7\pi}{20} + i \sin \frac{7\pi}{20} \right] \text{ ok } \frac{1}{2}$$

$$Z_3 = (\sqrt{18})^{\frac{1}{5}} \left[\cos \frac{15\pi}{20} + i \sin \frac{15\pi}{20} \right] \text{ ok } \frac{1}{2}$$

$$Z_4 = (\sqrt{18})^{\frac{1}{5}} \left[\cos \left(\frac{23\pi}{20} \right) + i \sin \left(\frac{23\pi}{20} \right) \right] \text{ ok } \frac{1}{2}$$

$$Z_5 = (\sqrt{18})^{\frac{1}{5}} \left[\cos \frac{31\pi}{20} + i \sin \frac{31\pi}{20} \right] \text{ ok } \frac{1}{2}$$

Page 20

5 (a) Given

Soln (SECTION B).

Page 21

$$\sin x + \sin y = a \quad \text{--- (i)}$$

$$\cos x + \cos y = b \quad \text{--- (ii)}$$

Required to show that

$$\cos^2\left(\frac{x-y}{2}\right) = \frac{1}{4}(a^2 + b^2)$$

from

$$\sin x + \sin y = 2 \sin \frac{1}{2}(x+y) \cos \frac{1}{2}(x-y)$$

$$\Rightarrow 2 \sin \frac{1}{2}(x+y) \cos \frac{1}{2}(x-y) = a \quad \text{--- (iii)}$$

Similarly

$$\cos x + \cos y = 2 \cos \frac{1}{2}(x+y) \cos \frac{1}{2}(x-y)$$

$$\Rightarrow 2 \cos \frac{1}{2}(x+y) \cos \frac{1}{2}(x-y) = b \quad \text{--- (iv)}$$

Squaring equations (iii) and (iv)

$$4 \sin^2 \frac{1}{2}(x+y) \cos^2 \frac{1}{2}(x-y) = a^2 \quad \text{--- (v)}$$

$$4 \cos^2 \frac{1}{2}(x+y) \cos^2 \frac{1}{2}(x-y) = b^2 \quad \text{--- (vi)}$$

Adding equations v and vi.

$$4 \sin^2 \frac{1}{2}(x+y) \cos^2 \frac{1}{2}(x-y) + 4 \cos^2 \frac{1}{2}(x+y) \cos^2 \frac{1}{2}(x-y) = a^2 + b^2$$

$$4 \cos^2 \frac{1}{2}(x-y) \left[\sin^2 \frac{1}{2}(x+y) + \cos^2 \frac{1}{2}(x+y) \right] = a^2 + b^2$$

$$4 \cos^2 \frac{1}{2}(x-y) = a^2 + b^2$$

(05)

$$\therefore \cos^2 \frac{1}{2}(x-y) = \frac{1}{4}(a^2 + b^2) \quad \text{hence shown}$$

5.

Soln. @

Page (22)

(b)

Given

$$\sin(X - 150^\circ) \cos(X + 150^\circ) = 0.5$$

$$\Rightarrow \frac{1}{2} [\sin((X - 150^\circ) - (X + 150^\circ)) + \sin((X - 150^\circ) + (X + 150^\circ))] = 0.5$$

$$\Rightarrow \frac{1}{2} [\sin(2X) + \sin(-300^\circ)] = 0.5$$

Multiplying both sides by 2

$$\Rightarrow [\sin(2X) + \sin(-300^\circ)] = 0.5 \times 2$$

$$\Rightarrow [\sin 2X - \sin 300^\circ] = 1.$$

$$\Rightarrow \sin 2X = 1 + \sin 300^\circ$$

$$\sin 2X = 1 - \frac{\sqrt{3}}{2}$$

$$\sin 2X = \frac{2 - \sqrt{3}}{2}$$

$$2X = \sin^{-1}\left(\frac{2 - \sqrt{3}}{2}\right)$$

$$2X = \sin^{-1}(0.13397)$$

$$X = 90^\circ n + (-1)^n (3.6993^\circ)$$

for $n = 0, 1, 2, 3, \dots$ The value of X are

$$X = 3.85^\circ,$$

$$X = 86.15^\circ,$$

$$X = 183.85^\circ \text{ and}$$

$$X = 266.15^\circ$$

(05)

5 (c)

Soln.

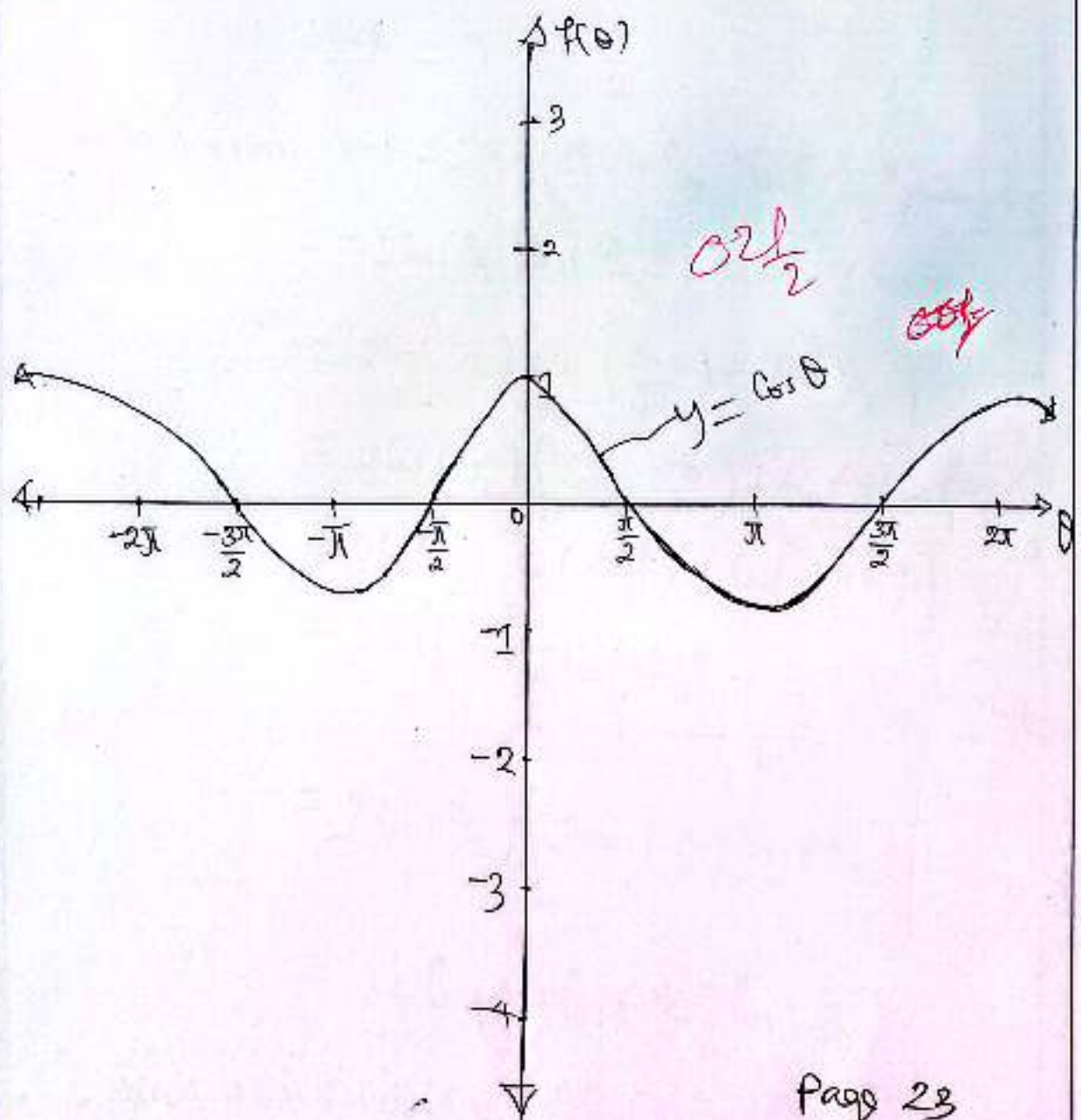
Given $f(\theta) = \cos \theta$, for $-2\pi \leq \theta \leq 2\pi$

Required to sketch the graph of $f(\theta)$

Prepare table of value

$\theta \frac{1}{2}$

θ	-2π	$-\frac{3\pi}{2}$	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$f(\theta)$	1	0	-1	0	1	0	-1	0	1



Page 23

5 (4).

Soln.

Page 24

for triangle ABC

Required to prove that $\sin A + \sin B + \sin C =$
 $4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}.$

Consider LHS

$$\sin A + \sin B + \sin C = 2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right) + 2 \sin \left(\frac{C}{2} \right) \cos \left(\frac{C}{2} \right)$$

But

$$A+B = 180^\circ - C$$

$$\Rightarrow \frac{A+B}{2} = 90^\circ - \frac{C}{2}$$

$$\Rightarrow \sin A + \sin B + \sin C = 2 \sin \left(90^\circ - \frac{C}{2} \right) \cos \left(\frac{A-B}{2} \right) + 2 \sin \frac{C}{2} \cos \frac{C}{2}$$

$$= 2 \cos \left(\frac{C}{2} \right) \cos \left(\frac{A-B}{2} \right) + 2 \sin \frac{C}{2} \cos \frac{C}{2}$$

$$= 2 \cos \left(\frac{C}{2} \right) \left[\cos \left(\frac{A-B}{2} \right) + \sin \frac{C}{2} \right]$$

$$= 2 \cos \left(\frac{C}{2} \right) \left[\cos \left(\frac{A-B}{2} \right) + \sin \left(90^\circ - \left(\frac{A+B}{2} \right) \right) \right]$$

$$= 2 \cos \left(\frac{C}{2} \right) \left[\cos \left(\frac{A-B}{2} \right) + \cos \left(\frac{A+B}{2} \right) \right]$$

$$= 2 \cos \left(\frac{C}{2} \right) \left[2 \cos \left(\frac{\frac{A+B}{2} + \frac{A-B}{2}}{2} \right) \cos \left(\frac{\frac{A+B}{2} - \frac{A-B}{2}}{2} \right) \right]$$

$$= 2 \cos \left(\frac{C}{2} \right) \left[2 \cos \left(\frac{A}{2} \right) \cos \left(\frac{B}{2} \right) \right]$$

$$= 4 \cos \left(\frac{A}{2} \right) \cos \left(\frac{B}{2} \right) \cos \left(\frac{C}{2} \right)$$

$$\therefore \sin A + \sin B + \sin C = 4 \cos \left(\frac{A}{2} \right) \cos \left(\frac{B}{2} \right) \cos \left(\frac{C}{2} \right)$$

(05 marks)

6 (a)

Soln.

Page 25

$$\frac{dy}{dx} + y = \sin x \quad \text{--- (1)}$$

compare with standard equation

$$\frac{dy}{dx} + P(x)y = Q(x) \text{ gives}$$

$$P(x) = 1$$

The integrating factor

$$I(x) = e^{\int P(x) dx}$$

$$= e^{\int dx}$$

$$I(x) = e^x$$

$$\frac{dy}{dx} + y = \sin x$$

Multiply by e^x both sides

$$e^x \frac{dy}{dx} + e^x y = e^x \sin x$$

$$\frac{d}{dx} (ye^x) = e^x \sin x$$

$$\int \frac{d}{dx} (ye^x) dx = \int e^x \sin x dx$$

(05 Marks)

$$y = \frac{1}{2} (\sin x - \cos x) + Ce^{-x}$$

∴ The general solution of the differential equation is

$$y = \frac{1}{2} (\sin x - \cos x) + Ce^{-x}$$

6

(b)

Soln.

Page 26

Required to find the values of p and q , for which the d.e

$$(y + x^3)dx + (px + qy^3)dy = 0 \text{ is exact}$$

$$M(x, y) = y + x^3$$

$$N(x, y) = px + qy^3$$

$$\frac{\partial M}{\partial y} = 1$$

$$\frac{\partial N}{\partial x} = p$$

$$\text{for exactness } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 1 = p$$

$$\therefore p = 1.$$

$q = \text{any arbitrary constant}$

$$f_{x,y} = y + x^3$$

$$f_x = y + x^3$$

$$F(x, y) = \int (y + x^3) dx$$

$$F(x, y) = xy + \frac{x^4}{4} + G(y)$$

(05 Marks)

$$f_y = x + g'(y)$$

$$= x + by^3$$

$$g'(y) = by^3$$

$$g(y) = \frac{b}{4} y^4$$

$$F(x, y) = xy + \frac{x^4}{4} + \frac{b}{4} y^4 \rightarrow C = xy + \frac{x^4}{4} + \frac{b}{4} y^4$$

6 (C) Given the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 2e^x - 10\sin x.$$

given that $y(0) = \frac{5}{2}$ and $y'(0) = \frac{19}{2}$.

The characteristic equation of the homogeneous part is given by

$$\lambda^2 - 2\lambda - 3 = 0$$

$$\Rightarrow (\lambda + 1)(\lambda - 3) = 0$$

$$\lambda_1 = -1 \text{ and } \lambda_2 = 3$$

$$y_c = c_1 e^{-x} + c_2 e^{3x}.$$

$$\text{let } y_p = \lambda e^x + A \sin x + B \cos x$$

$$\Rightarrow y_p' = \lambda e^x + A \cos x - B \sin x$$

$$y_p'' = \lambda e^x - A \sin x - B \cos x$$

Substituting y_p'' , y_p' and y_p in the given d.e

$$-4\lambda e^x + (-4A + 2B)\sin x + (-4B - 2A)\cos x = 2e^x - 10\sin x$$

$$-4\lambda e^x = 2e^x \quad \text{--- (i)}$$

$$4A + 2B = -10 \quad \text{--- (ii)}$$

$$-4B - 2A = 0 \quad \text{--- (iii)}$$

(03 marks)

$$\lambda = -\frac{1}{2}, A = 2 \text{ and } B = -1.$$

$$\therefore y_p = -\frac{1}{2}e^x + 2\sin x - \cos x.$$

6 (c) General solution is

$$y = y_c + y_p$$

$$y = c_1 e^{-x} + c_2 e^{3x} - \frac{1}{2} e^x + 2 \sin x - \cos x$$

Given $x=0$ and $y = \frac{5}{2}$

$$c_1 + c_2 = 4 \quad \text{--- (i)}$$

$$y' = -c_1 e^{-x} + 3c_2 e^{3x} - \frac{1}{2} e^x + 2 \cos x + \sin x$$

$x=0$ $y' = \frac{19}{2}$

$$-c_1 + 3c_2 = 8 \quad \text{--- (ii)}$$

Solving eqn (i) and (ii)

$$c_1 = 1 \text{ and } c_2 = 3$$

(02 marks)

$$\therefore Y = e^{-x} + 3e^{3x} - \frac{1}{2} e^x + 2 \sin x - \cos x$$

(d)

Soln.

Let N be the number of inhabitants present at time t

$$\frac{dN}{dt} \propto N$$

$$\propto \frac{1}{2}$$

$\Rightarrow \frac{dN}{dt} = kN$, where k is the constant of proportionality

$$\int \frac{dN}{N} = k \int dt$$

$$\ln |N| = kt + C$$

(01)

6 (d) $\Rightarrow N = Ae^{kt}$ where $e^c = A$
 At $t=0 \Rightarrow N = N_0$

$$N_0 = Ae^{k(0)} \quad \text{001/2}$$

$$N_0 = A$$

$$N = N_0 e^{kt} \quad \text{001/2}$$

when $t = 50$, $N = 2N_0$

$$2N_0 = N_0 e^{k(50)}$$

$$2 = e^{50k}$$

$$\ln 2 = \ln e^{50k}$$

$$50k = \ln 2$$

$$k = \frac{1}{50} \ln 2 \quad \text{(6)}$$

Required to find the time t when

$$N = 3N_0$$

$$3N_0 = N_0 e^{\left(\frac{1}{50} \ln 2\right)t}$$

$$\ln 3 = \frac{t}{50} \ln 2 \quad \text{001/2}$$

$$\Rightarrow t = \frac{50 \ln 3}{\ln 2}$$

$$\text{Hence } t = 79.24812504 \quad \text{001/2}$$

$\therefore$ The population will be three times the original after 79.2 years. 001/2

7

(a)

SECTION B

Page 30

Soln.

Given the line $ax + by + c = 0$ as a tangent to the parabola $x^2 = 4ay$.

Required to find the equation connecting a, b and c from

$$ax + by + c = 0 \quad \text{--- (i)}$$

$$x^2 = 4ay \quad \text{--- (ii)}$$

$$\Rightarrow 4ay = x^2$$

$$y = \frac{x^2}{4a} \quad \text{--- (iii)}$$

Substituting eqn (iii) into (i)

$$ax + b\left(\frac{x^2}{4a}\right) + c = 0$$

$$4a^2x + bx^2 + 4ac = 0$$

$$bx^2 + 4a^2x + 4ac = 0$$

$$\Rightarrow a = b, \quad b = 4a^2, \text{ and } c = 4ac$$

For a line to be tangent to the parabola then

$$b^2 = 4ac$$

$$(4a^2)^2 = 4(b)(4ac)$$

$$16a^4 = 16abc$$

$$bc = a^3$$

(05 marks)

$\therefore$ The equation connecting a, b , and c is $bc = a^3$

7

(b)

Soln.

Page (31)

Given the equation $18x^2 + 12y^2 - 144x + 48y + 120 = 0$
 Required to write its equation in standard form
 of an ellipse as $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

from
 $18x^2 + 12y^2 - 144x + 48y + 120 = 0$

$$18x^2 - 144x + 12y^2 + 48y = -120$$

$$(18x^2 - 144x + A) + (12y^2 + 48y + B) = -120 + A + B$$

For a quadratic to be perfect square

$$18x^2 - 144x + A = 0$$

$$b^2 = 4ac$$

$$(-144)^2 = 4(18)A$$

$$20736 = 72A$$

$$\Rightarrow A = 288$$

Also

$$12y^2 + 48y + B = 0$$

$$(48)^2 = 4(12)B$$

$$2304 = 48B$$

$$48 = B$$

$$\Rightarrow B = 48$$

(0.5 marks)

$$(18x^2 - 144x + 288) + (12y^2 + 48y + 48) = -120 + 288 + 48$$

$$18(x^2 - 8x + 16) + 12(y^2 + 4y + 4) = 216$$

$$18(x-4)^2 + 12(y+2)^2 = 216$$

Dividing both sides by 216

$$\frac{(x-4)^2}{12} + \frac{(y+2)^2}{18} = 1$$

7 (c)

Solve

Page 32

Given the polar equation as

$$r = \frac{4}{3+5\sin\theta} \quad \text{--- (i)}$$

Required to find the cartesian equation from

$$x = r\cos\theta \quad \text{--- (ii)}$$

$$y = r\sin\theta \quad \text{--- (iii)}$$

$$\sin\theta = \frac{y}{r} \quad \text{--- (iv)}$$

$$r = \frac{4}{3+5\left(\frac{y}{r}\right)}$$

$$r = \frac{4}{\frac{3r+5y}{r}}$$

$$r = \frac{4r}{3r+5y}$$

Divide by r both sides:

$$\frac{r}{r} = \frac{4r}{3r+5y} \times \frac{1}{r}$$

$$\frac{1}{1} = \frac{4}{3r+5y}$$

$$3r+5y = 4$$

$$3r = 4-5y$$

Squaring both sides

$$9r^2 = (4-5y)^2$$

$$9r^2 = 16-40y+25y^2$$

$$\text{but } r^2 = x^2 + y^2$$

$$9x^2 + 9y^2 = 16 - 40y + 25y^2$$

$$9x^2 - 16y^2 + 40y - 16 = 0$$

$\therefore$ The cartesian equation required is

$$9x^2 - 16y^2 + 40y - 16 = 0$$

(05 marks)

7 (8)

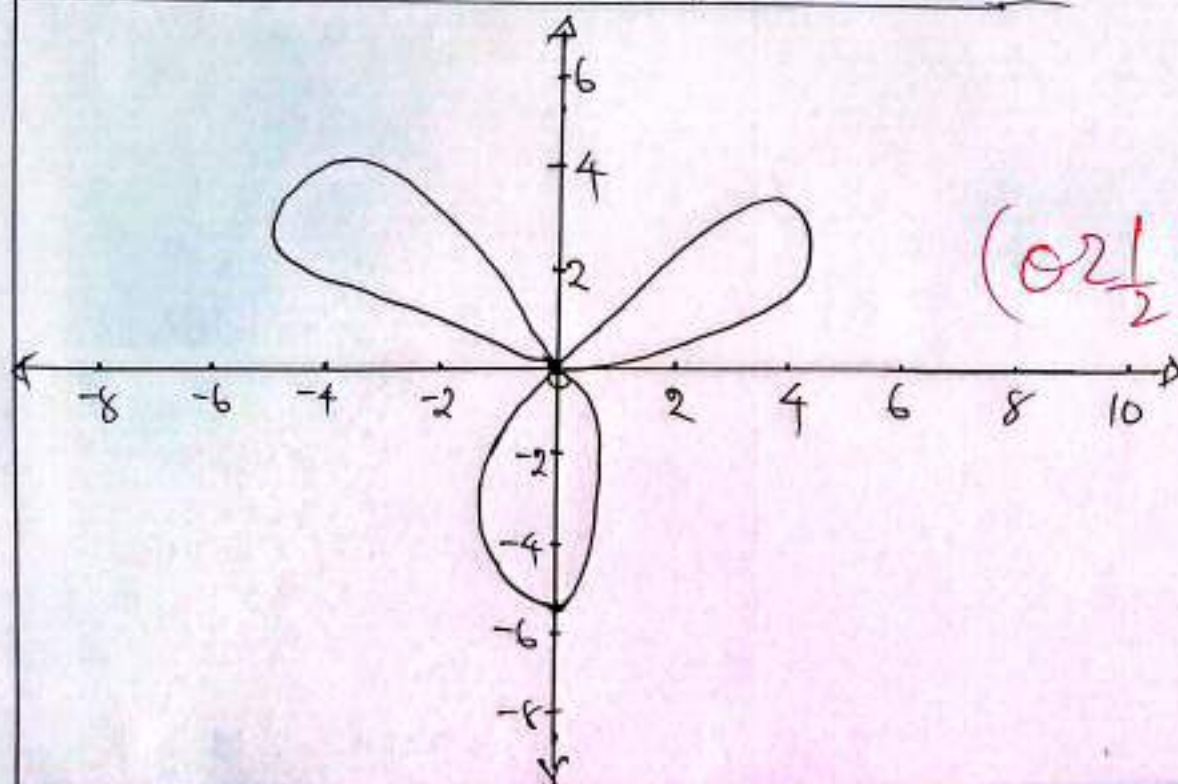
Soln

Page 33

Given the equation $r = 5 \sin 3\theta$ Prepare table of values of the polar equation
 $r = 5 \sin 3\theta$

θ	$r = 5 \sin 3\theta$	(r, θ)
0°	0	$(0, 0^\circ)$
30°	5	$(5, 30^\circ)$
60°	0	$(0, 60^\circ)$
90°	-5	$(-5, 90^\circ)$
120°	0	$(0, 120^\circ)$
150°	5	$(5, 150^\circ)$
180°	0	$(0, 180^\circ)$
210°	-5	$(-5, 210^\circ)$
240°	0	$(0, 240^\circ)$
270°	5	$(5, 270^\circ)$
300°	0	$(0, 300^\circ)$
330°	-5	$(-5, 330^\circ)$
360°	0	$(0, 360^\circ)$

(02 1/2)



(02 1/2)